MBA · DIGITAL BUSINESS VALUATION

Week 2: Discounted Cash Flow Valuation

Foundations from Aswath Damodaran's Investment Valuation

Source: A. Damodaran, *Investment Valuation: Tools and Techniques for Determining the Value of Any Asset*, 3rd Edition (University Edition) — Chapters 2 - 9

https://pages.stern.nyu.edu/~adamodar/New_Home_Page/valuationtools.html#valtools

Course Instructor: Oktofa Yudha Sudrajad, PhD

One Goal, Three Different Routes

1

Discounted Cash Flow

Value = present value of the expected future cash flows on the asset.

2

Relative Valuation

Value is derived from how comparable assets are priced on a common variable — earnings, book value, or sales.

3

Contingent Claim (Option)

Uses option-pricing models to value assets whose payoff depends on an event occurring or not (“real options”).

These are not mutually exclusive. Most real-world valuations are relative valuations — but DCF is the foundation on which the other two approaches are built.

Damodaran, Ch.2, pp.11, 25

The Basis: Present Value of Expected Cash Flows

$$\text{Value} = \Sigma \ CF_t / (1 + r)^t$$

n = life of the asset · CF_t = cash flow in period t · r = discount rate reflecting risk

- Cash flows vary by asset: dividends for stocks, coupons and face value for bonds, after-tax cash flows for a real project.
- The discount rate reflects the riskiness of the estimated cash flows — higher for riskier assets, lower for safer ones.

Damodaran, Ch.2, p.12

The Risk Continuum

1. Riskless zero-coupon bond: discount the guaranteed cash flow at the riskless rate.
2. Corporate bonds: coupons carry default risk; discount at a rate reflecting that risk.
3. Equities: cash flows carry substantial uncertainty; discount at a rate reflecting it.

Underpinning: DCF seeks intrinsic value — what an unbiased analyst would attach to the firm given correct cash-flow and discount-rate estimates.

Basis & Categorization

The Basis

- Value is derived from the pricing of comparable assets, standardized by a common variable: earnings (P/E), book value (P/BV), revenues (P/S), or EV/EBITDA.
- Relies on the market being right on average, even if it misprices individual stocks — and on those errors correcting over time.
- A fundamentally different premise from DCF's search for intrinsic value.

Damodaran, Ch.2, pp.19–20

Fundamentals vs. Comparables

Some analysts extract multiples from DCF fundamentals; most simply compare a firm's multiple to similar firms — often without stating the assumptions this requires.

Cross-Sectional vs. Time Series

Cross-sectional: compare a firm's multiple to other firms today. Time series: compare a mature firm's current multiple to its own history — assuming its fundamentals haven't changed.

Applicability & Limitations

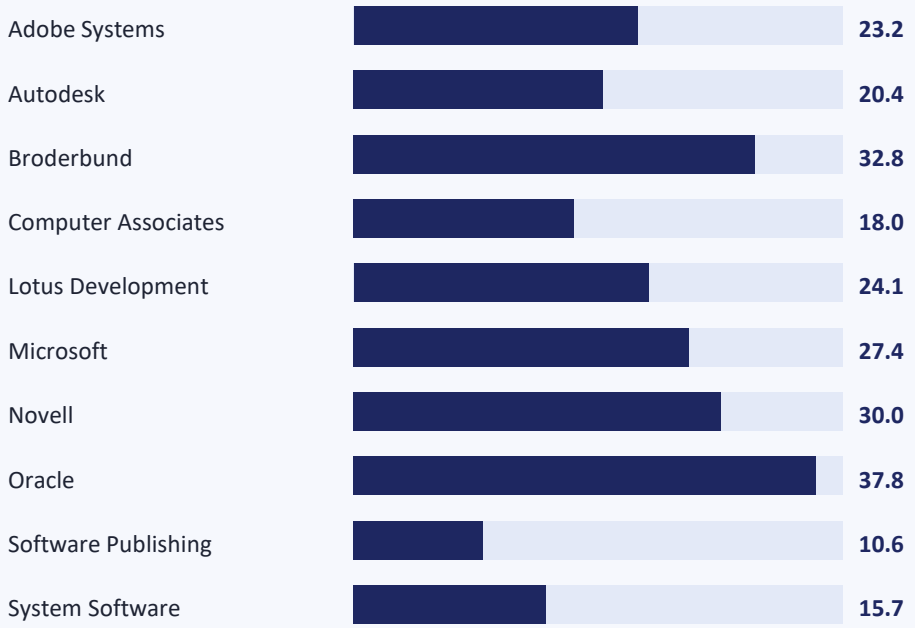
- Strength: simple and fast — useful when many good comparables trade in liquid, correctly-priced markets.
- Weakness: “no two firms are exactly alike” — the choice of comparables is inherently subjective.
- Also inherits the market’s own mispricing of the whole comparable group — a risk DCF is less exposed to.

Damodaran, Ch.2, pp.21, 23

Illustration — P/E of 10 “Comparable” Software Firms

- Full sample of 10 firms → average P/E = 24.0
- Drop the 2 lowest (10.6, 15.7) → average rises to 27
- Drop the 2 highest (32.8, 37.8) → average falls to 21

Range of P/E Multiples Across 10 “Comparable” Firms



Average = 24.0× (range narrows to 21–27× depending on which firms are kept in the sample)

Basis & the Payoff Diagram

The Basis

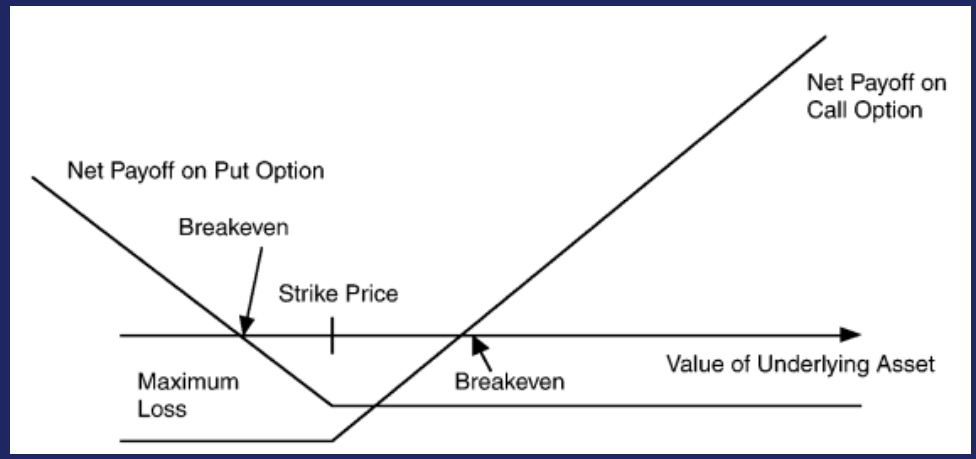
- Some assets are worth more than the PV of their expected cash flows because their payoff is contingent on an event occurring — or not.
- Valued using option-pricing models (Black-Scholes, 1972; the binomial model) developed for traded options and extended to real assets.
- A call option pays off if the underlying value exceeds a strike price; a put option pays off if it falls below one.

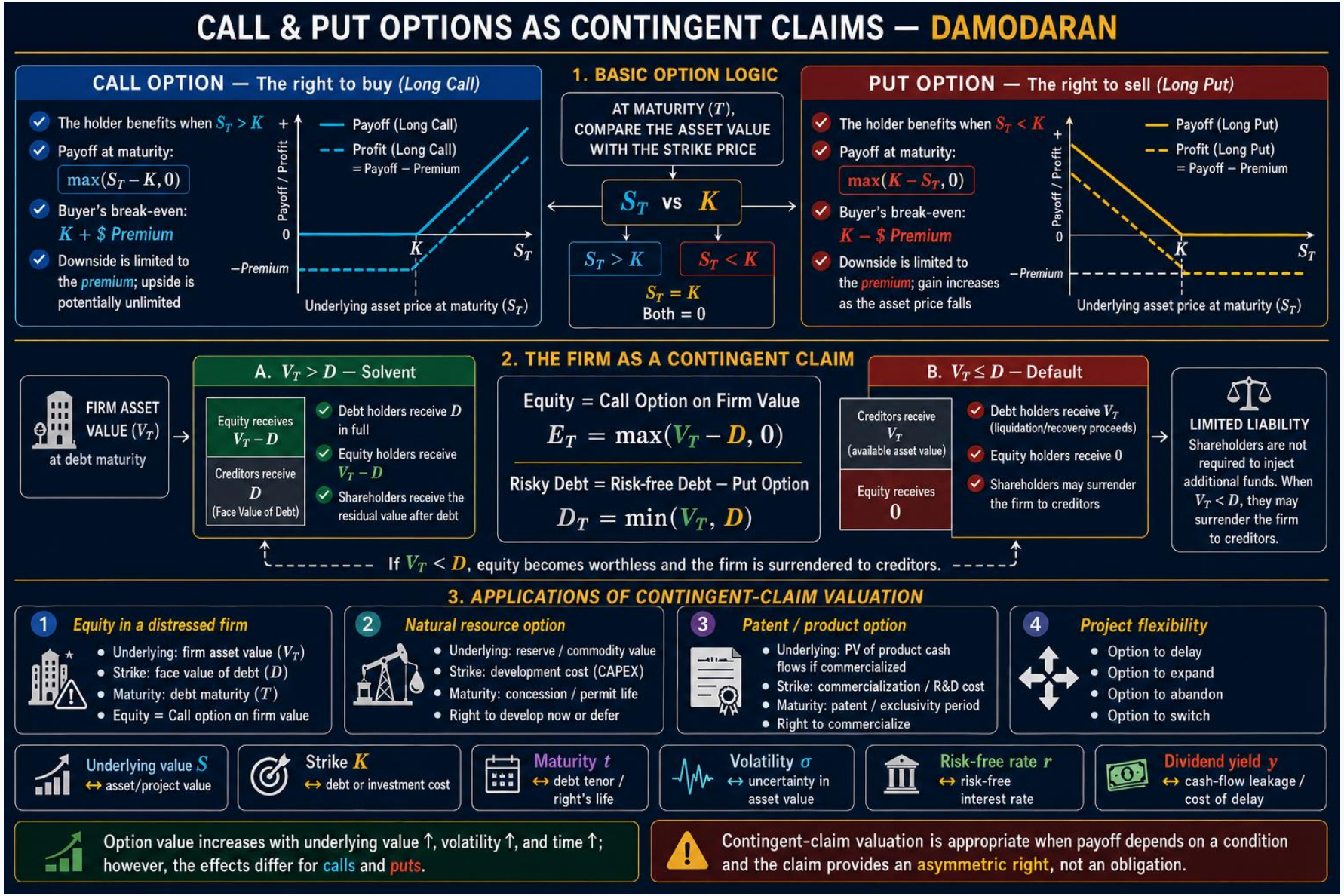
Damodaran, Ch.2, pp.23–24

Figure Payoff on Call & Put Options

Underpinning: DCF tends to understate value when a payoff depends on a future event — e.g., an undeveloped oil reserve is worth more than a straight cash-flow forecast suggests, because it will only be developed if oil prices rise.

Call payoff Put payoff




CALL & PUT OPTIONS AS CONTINGENT CLAIMS — DAMODARAN
CALL OPTION — The right to buy (Long Call)
The holder benefits when $S_T > K$
Payoff at maturity:
$\max(S_T - K, 0)$
Buyer's break-even:
$K + \$ Premium$
Downside is limited to the premium; upside is potentially unlimited
Payoff / Profit
+
0
−Premium
K
S_T
Payoff (Long Call)
Profit (Long Call)
= Payoff − Premium
Underlying asset price at maturity (S_T)
1. BASIC OPTION LOGIC
AT MATURITY (T), COMPARE THE ASSET VALUE WITH THE STRIKE PRICE
S_T vs K
$S_T > K$
$S_T < K$
$S_T = K$
Both = 0
PUT OPTION — The right to sell (Long Put)
The holder benefits when $S_T < K$
Payoff at maturity:
$\max(K - S_T, 0)$
Buyer's break-even:
$K - \$ Premium$
Downside is limited to the premium; gain increases as the asset price falls
Payoff / Profit
+
0
−Premium
K
S_T
Payoff (Long Put)
Profit (Long Put)
= Payoff − Premium
Underlying asset price at maturity (S_T)
FIRM ASSET VALUE (V_T)
at debt maturity
A. $V_T > D$ — Solvent
Equity receives $V_T - D$
Creditors receive D (Face Value of Debt)
Debt holders receive D in full
Equity holders receive $V_T - D$
Shareholders receive the residual value after debt
2. THE FIRM AS A CONTINGENT CLAIM
Equity = Call Option on Firm Value
$E_T = \max(V_T - D, 0)$
Risky Debt = Risk-free Debt − Put Option
$D_T = \min(V_T, D)$
B. $V_T \leq D$ — Default
Creditors receive V_T (available asset value)
Equity receives 0
Debt holders receive V_T (liquidation/recovery proceeds)
Equity holders receive 0
Shareholders may surrender the firm to creditors
LIMITED LIABILITY
Shareholders are not required to inject additional funds. When $V_T < D$, they may surrender the firm to creditors.
If $V_T < D$, equity becomes worthless and the firm is surrendered to creditors.
3. APPLICATIONS OF CONTINGENT-CLAIM VALUATION
1 Equity in a distressed firm
Underlying: firm asset value (V_T)
Strike: face value of debt (D)
Maturity: debt maturity (T)
Equity = Call option on firm value
2 Natural resource option
Underlying: reserve / commodity value
Strike: development cost (CAPEX)
Maturity: concession / permit life
Right to develop now or defer
3 Patent / product option
Underlying: PV of product cash flows if commercialized
Strike: commercialization / R&D cost
Maturity: patent / exclusivity period
Right to commercialize
4 Project flexibility
Option to delay
Option to expand
Option to abandon
Option to switch
Underlying value S ↔ asset/project value
Strike K ↔ debt or investment cost
Maturity t ↔ debt tenor / right's life
Volatility σ ↔ uncertainty in asset value
Risk-free rate r ↔ risk-free interest rate
Dividend yield y ↔ cash-flow leakage / cost of delay
Option value increases with underlying value ↑, volatility ↑, and time ↑; however, the effects differ for calls and puts.
Contingent-claim valuation is appropriate when payoff depends on a condition and the claim provides an asymmetric right, not an obligation.

Applicability: Real Options

Direct Examples

- LEAPS — long-term equity options on traded stocks.
- Warrants — long-term call options issued by firms.
- Contingent value rights — protect stockholders against price declines.

"Real Option" Examples

- A patent — a call option on a product; the investment outlay is the strike price, the patent life is the time to expiration.
- An undeveloped natural-resource reserve.
- Equity in a heavily levered firm — a call option on firm value, with the face value of debt as the strike price.

Damodaran, Ch.2, pp.24–25

Limitations

- Assumptions such as constant variance and no dividends are far harder to defend over long option lifetimes.
- For non-traded underlying assets, inputs (value, variance) cannot be pulled from financial markets and must be estimated.
- As a result, values from these applications carry much more estimation error than standard, short-term listed options.

Comparing the Three Approaches

	Discounted Cash Flow	Relative Valuation	Contingent Claim
Basis	PV of expected cash flows on the asset	Pricing of comparable assets on a common variable	Option-pricing models for event-contingent payoffs
Best suited for	Stable, cash-flow-generating firms with an estimable risk profile	Firms with many good, liquid comparables	Assets with optionality — patents, reserves, distressed equity
Key limitation	Hard to apply to distressed, cyclical, or private firms	Comparable selection is subjective; inherits market mispricing	Assumptions strain over long, non-traded time horizons

Not mutually exclusive — a complete valuation often draws on more than one approach.

Damodaran, Ch.2, p.25 (Conclusion)

Why This Matters for Digital Business Valuation

Earnings Come Later

Digital and tech businesses often post negative or highly volatile near-term earnings — pushing straight into DCF's hardest applicability zone (Myth 3; Ch.2 Applicability).

Comparables Are Everywhere — and Risky

P/S, EV/GMV, and EV/subscriber multiples are common for digital firms — but Illustration 2.2's lesson about comparable-selection bias applies directly.

Growth Looks Like an Option

Platform network effects and unproven business models often behave like real options — the same logic used for patents and undeveloped reserves.

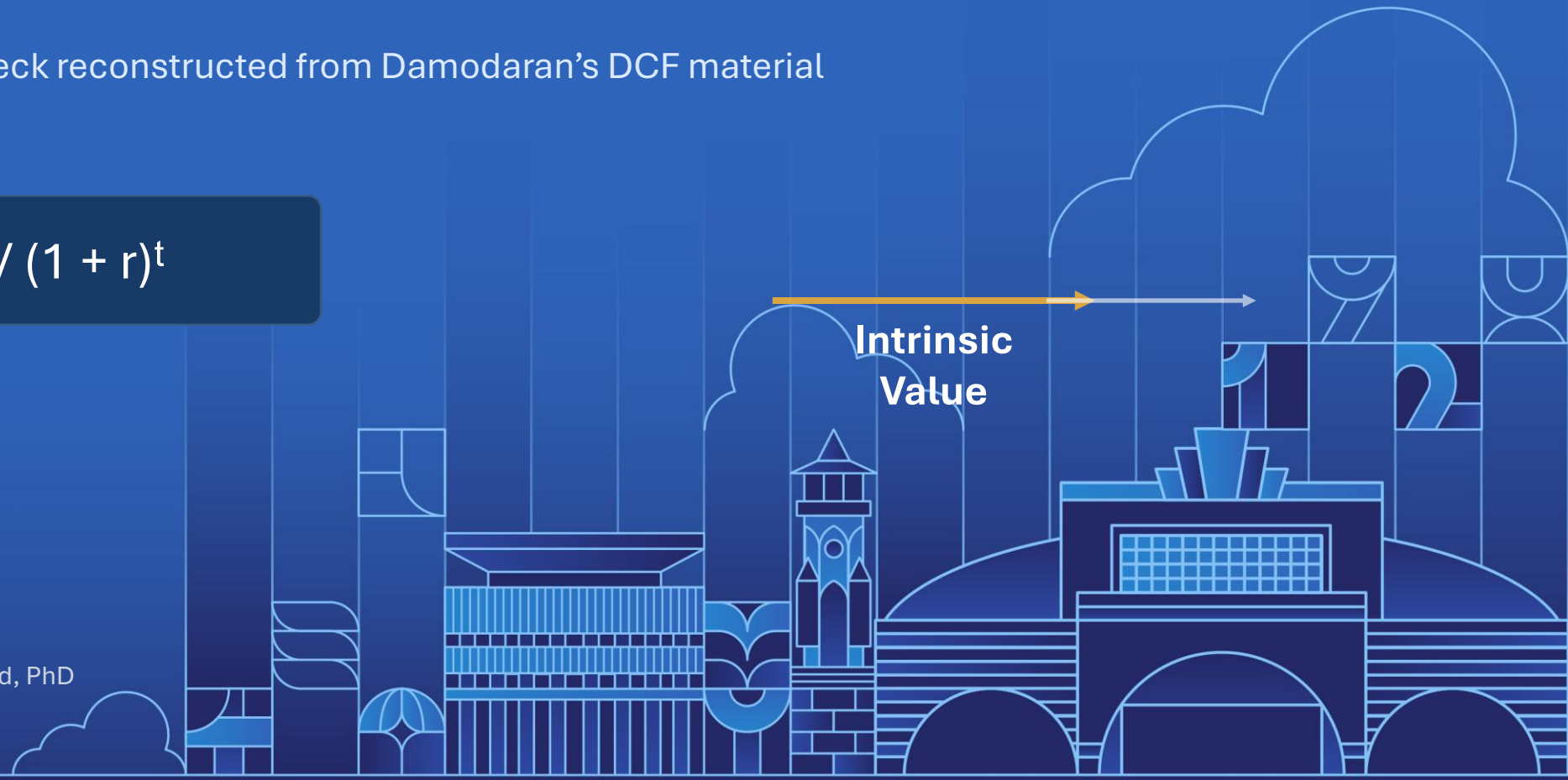
Precision Is Not the Point

Myths 2 and 3 — timelessness and precision — are especially relevant: digital valuations age fast and are rarely precise. A defensible range, not a point estimate, is the honest output.

Discounted Cash Flow Valuation

A comprehensive teaching deck reconstructed from Damodaran's DCF material

$$\text{Value} = \Sigma \, CF_t / (1 + r)^t$$



Course Instructor: Oktofa Yudha Sudrajad, PhD
Session · SBM ITB

Learning Outcomes

1 Explain the DCF logic

Interpret value as the present value of expected future cash flows, with time and risk priced explicitly.

2 Build consistent models

Match cash flows with discount rates: FCFE with cost of equity, FCFF with WACC.

3 Estimate core inputs

Translate accounting data into cash flows; estimate discount rates, growth, and terminal value.

4 Choose the right DCF model

Select stable, two-stage, three-stage, or n-stage models for different company life cycles.

5 Diagnose valuation errors

Identify mismatches, unrealistic terminal growth, overlooked reinvestment, and loose-end claims.

6 Apply DCF to digital firms

Handle negative earnings, changing margins, intangible reinvestment, and option-like growth.

Roadmap

1	**DCF Foundation**	present value · equity vs firm · consistency
2	**Discount Rates**	risk-free rate · beta · ERP · WACC
3	**Cash Flows**	FCFF · FCFE · accounting adjustments
4	**Growth & Reinvestment**	historical · analyst · fundamentals
5	**Terminal Value**	stable growth · closure · sensitivity
6	**Model Selection**	special situations · digital firms · checklist

PART 1

DCF Foundation

What DCF is and what it is not

DCF



CORE IDEA

DCF in One Sentence

$$\text{Value} = \sum_{t=1}^{n} CF_t / (1 + r)^t$$

Cash flow

Expected cash flow in each period. Different assets imply different cash flows: dividends, FCFE, FCFF, coupons, project cash flows.

Discount rate

The required return for bearing the risk of those cash flows. Riskier cash flows require higher discount rates.

Life / horizon

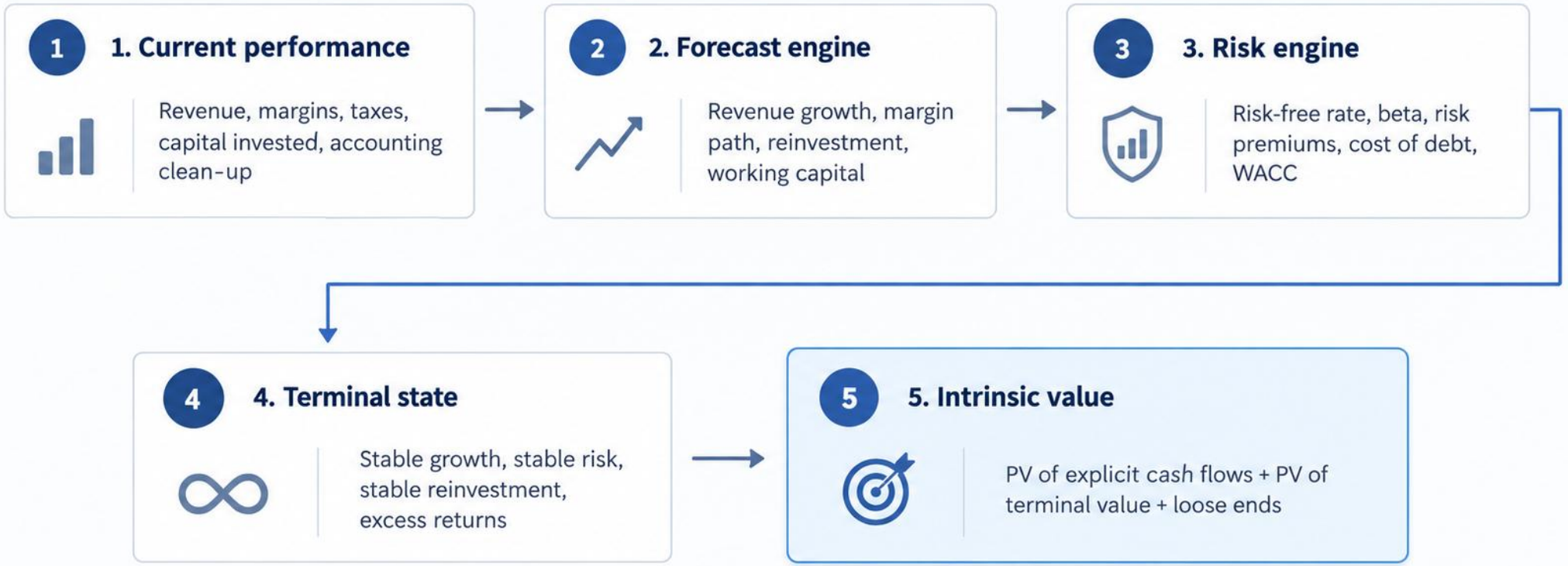
The number of periods over which explicit cash flows are forecast, followed by a terminal value when appropriate.

Two propositions: assets need positive expected cash flows at some point, and earlier cash flows are more valuable unless later growth compensates.

The Full DCF Architecture

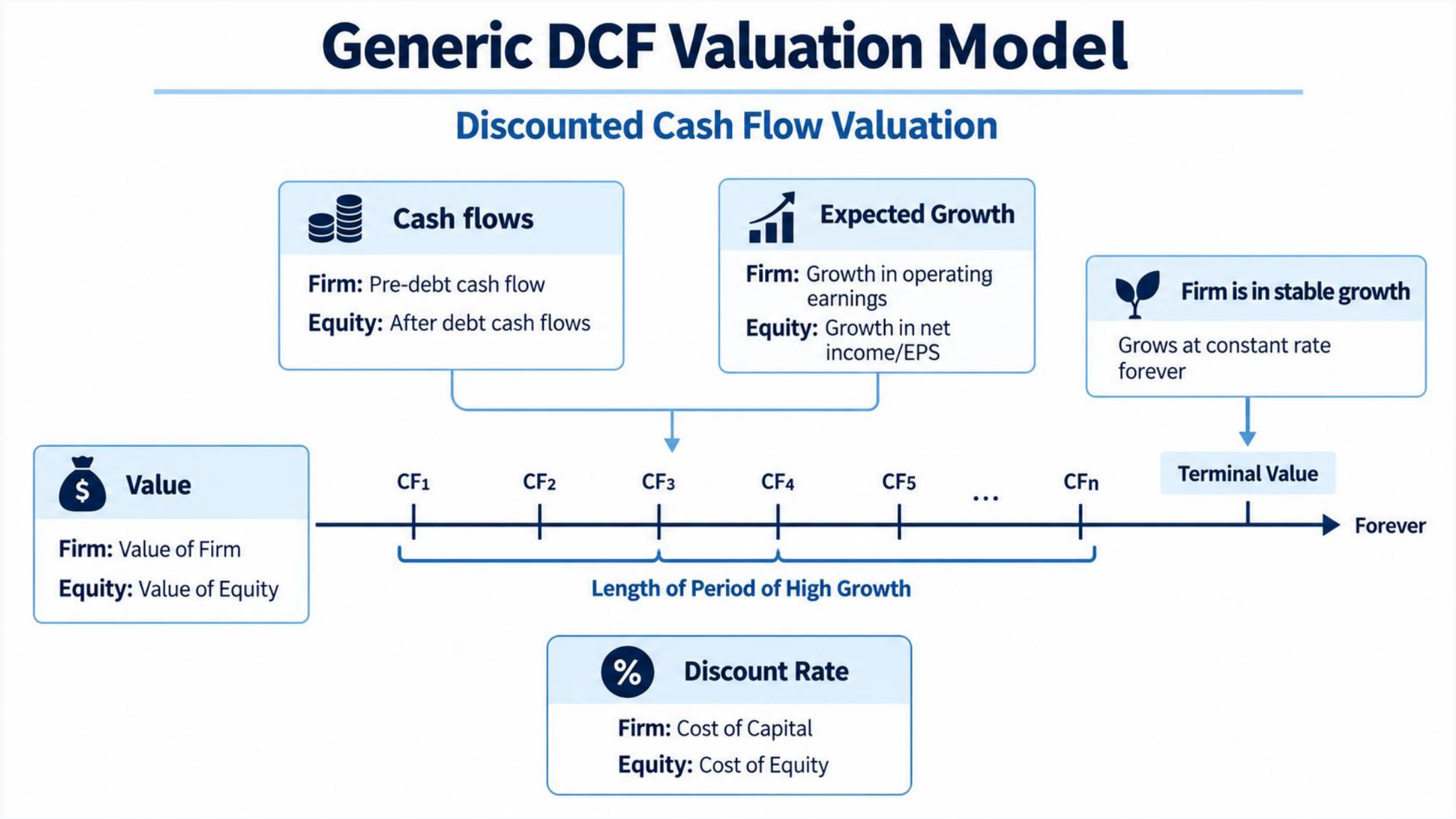
AN INTEGRATED FRAMEWORK FOR VALUATION

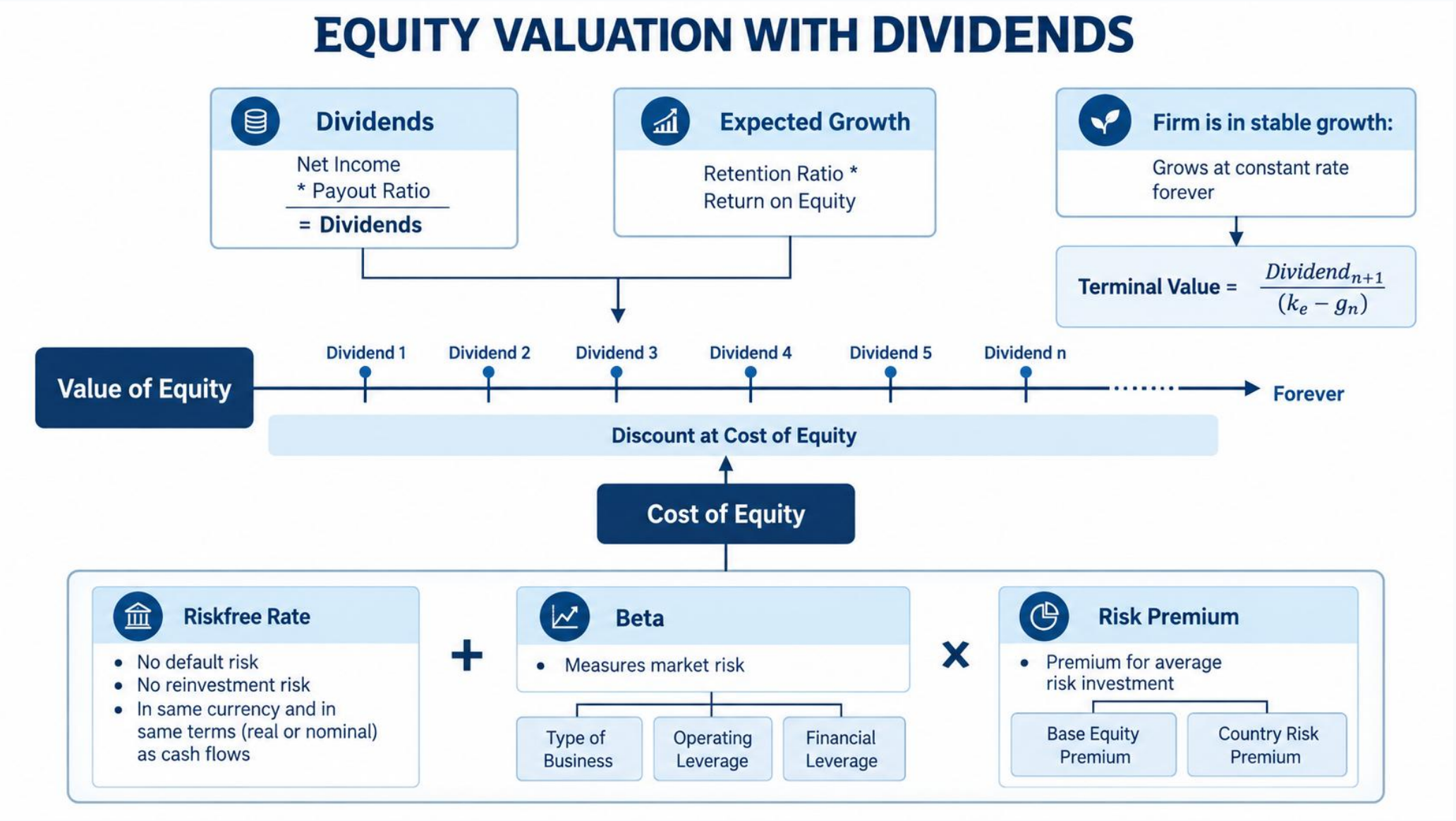
FROM BUSINESS REALITY TO INTRINSIC VALUE

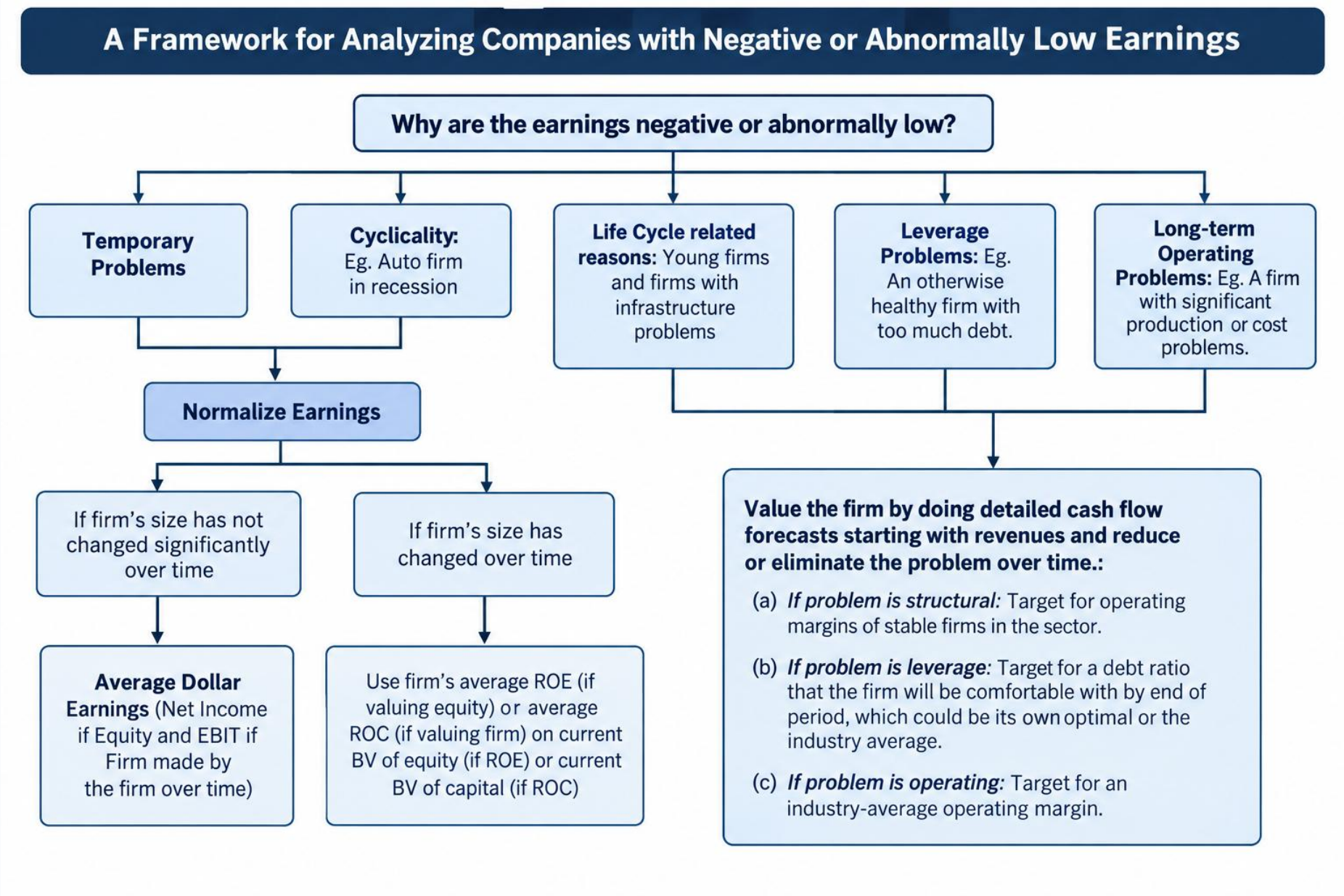


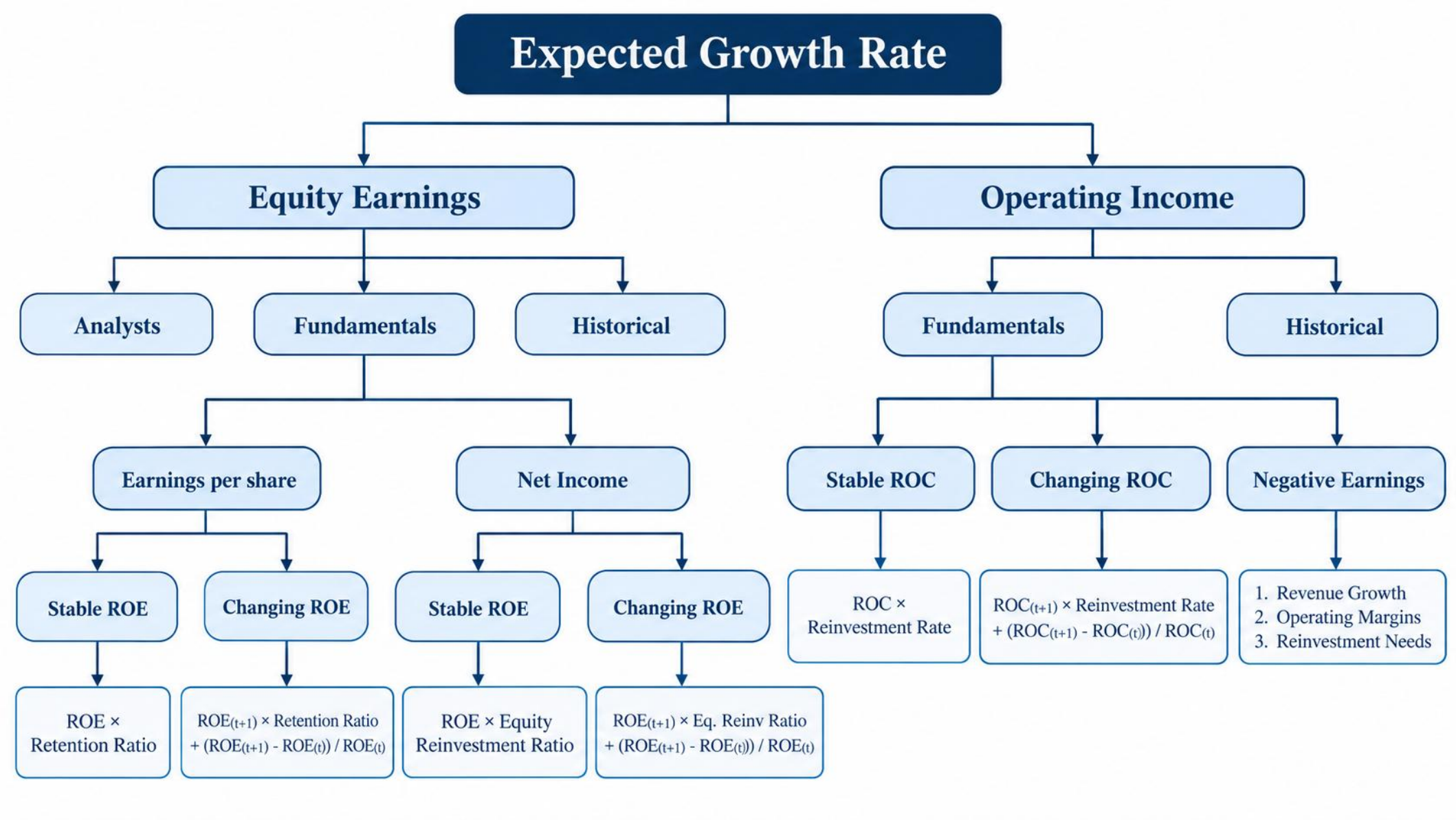
DISCOUNTED CASH FLOW ANALYSIS

BETTER ANALYSIS. BRIGHTER DECISIONS.


Generic DCF Valuation Model
Discounted Cash Flow Valuation
Cash flows
Firm: Pre-debt cash flow
Equity: After debt cash flows
Expected Growth
Firm: Growth in operating earnings
Equity: Growth in net income/EPS
Firm is in stable growth
Grows at constant rate forever
Value
Firm: Value of Firm
Equity: Value of Equity
CF1
CF2
CF3
CF4
CF5
...
CFn
Terminal Value
Forever
Length of Period of High Growth
Discount Rate
Firm: Cost of Capital
Equity: Cost of Equity


EQUITY VALUATION WITH DIVIDENDS
Dividends
Net Income
* Payout Ratio
= Dividends
Expected Growth
Retention Ratio *
Return on Equity
Firm is in stable growth:
Grows at constant rate forever
Terminal Value = $\frac{Dividend_{n+1}}{(k_e - g_n)}$
Value of Equity
Dividend 1
Dividend 2
Dividend 3
Dividend 4
Dividend 5
Dividend n
Forever
Discount at Cost of Equity
Cost of Equity
Riskfree Rate
No default risk
No reinvestment risk
In same currency and in same terms (real or nominal) as cash flows
+
Beta
Measures market risk
Type of Business
Operating Leverage
Financial Leverage
×
Risk Premium
Premium for average risk investment
Base Equity Premium
Country Risk Premium


A Framework for Analyzing Companies with Negative or Abnormally Low Earnings
Why are the earnings negative or abnormally low?
Temporary Problems
Cyclicality: Eg. Auto firm in recession
Life Cycle related reasons: Young firms and firms with infrastructure problems
Leverage Problems: Eg. An otherwise healthy firm with too much debt.
Long-term Operating Problems: Eg. A firm with significant production or cost problems.
Normalize Earnings
If firm's size has not changed significantly over time
If firm's size has changed over time
Average Dollar Earnings (Net Income if Equity and EBIT if Firm made by the firm over time)
Use firm's average ROE (if valuing equity) or average ROC (if valuing firm) on current BV of equity (if ROE) or current BV of capital (if ROC)
Value the firm by doing detailed cash flow forecasts starting with revenues and reduce or eliminate the problem over time.:
(a) If problem is structural: Target for operating margins of stable firms in the sector.
(b) If problem is leverage: Target for a debt ratio that the firm will be comfortable with by end of period, which could be its own optimal or the industry average.
(c) If problem is operating: Target for an industry-average operating margin.


Expected Growth Rate
Equity Earnings
Operating Income
Analysts
Fundamentals
Historical
Fundamentals
Historical
Earnings per share
Net Income
Stable ROC
Changing ROC
Negative Earnings
Stable ROE
Changing ROE
Stable ROE
Changing ROE
ROC × Reinvestment Rate
ROC(t+1) × Reinvestment Rate + (ROC(t+1) - ROC(t)) / ROC(t)
1. Revenue Growth
2. Operating Margins
3. Reinvestment Needs
ROE × Retention Ratio
ROE(t+1) × Retention Ratio + (ROE(t+1) - ROE(t)) / ROE(t)
ROE × Equity Reinvestment Ratio
ROE(t+1) × Eq. Reinv Ratio + (ROE(t+1) - ROE(t)) / ROE(t)

Equity Valuation vs Firm Valuation

Equity route

Discount cash flows left after debt payments and reinvestment.

Value of equity = PV(FCFE or dividends) at k_e

Firm route

Discount cash flows before debt payments but after taxes and reinvestment.

Value of firm = PV(FCFF) at WACC

Equity Value = Firm Value − Debt − Other Non-Equity Claims

The valuation route is a choice of perspective, not a choice of value. Consistency determines whether both routes converge.

NEVER MIX AND MATCH CASH FLOWS AND RATES

First Principle of Valuation

Cash flow being discounted	Correct discount rate	Output
Dividends / FCFE	Cost of equity (k_e)	Equity value
FCFF	WACC	Firm value
Real cash flows	Real discount rate	Real valuation
Nominal cash flows	Nominal discount rate	Nominal valuation
IDR cash flows	IDR discount rate	IDR value
USD cash flows	USD discount rate	USD value

Classic errors

FCFE discounted at WACC → equity value biased upward.
FCFF discounted at cost of equity → firm value biased downward.

A Numeric Consistency Check

Method	Cash flow	Discount rate	Value result	Interpretation
Correct equity route	CF to equity	13.625% k_e	$1,073	Equity value
Correct firm route	CF to firm	9.94% WACC	$1,873 − $800 = $1,073	Same equity value
Error 1	CF to equity	WACC	$1,248	Overstated equity
Error 2	CF to firm	k_e	$1,613 firm; $813 equity	Understated firm/equity
Error 3	CF to firm	k_e and forget debt	$1,613 equity	Severely overstated equity

Takeaway: the numbers may look precise, but the logic fails immediately when the cash-flow claimholder and discount-rate claimholder are not the same.

WORKFLOW FROM INPUTS TO VALUE

DCF Valuation: The Five Steps

1 **Estimate discount rate(s)**

k_e for equity; WACC for firm; match currency and inflation.

2 **Measure current cash flows**

FCFE, FCFF, or dividends; adjust accounting where needed.

3 **Forecast future cash flows**

Link growth to margins, reinvestment, and returns.

4 **Estimate terminal value**

When the firm reaches stable growth.

5 **Choose the right DCF model**

Stable, 2-stage, 3-stage, or n-stage.

PART 2

Discount Rates

How risk, currency, and financing enter valuation

DCF



Discount Rate Consistency

Claimholder consistency

Equity cash flows → k_e. Firm cash flows → WACC.

Currency consistency

Cash flows in rupiah require a rupiah discount rate; USD cash flows require USD rates.

Inflation consistency

Nominal cash flows → nominal rates. Real cash flows → real rates.

Time consistency

Rates may vary across time when leverage, risk, or macro conditions change.

Risk consistency

Discount rate should reflect systematic risk and default risk relevant to the cash flow.

NOT EVERY GOVERNMENT RATE IS RISK-FREE

The Risk-Free Rate

Risk-free asset: actual return = expected return

Two requirements

No default risk
No reinvestment risk

Practical proxy

Match the duration of the risk-free rate to the horizon of the cash flows, usually long-term.

Emerging markets

Local government bonds may embed default risk; use default spread adjustments or value in a stable currency.

Rule: The risk-free rate is chosen for the currency of the cash flows, not the country where the company is incorporated.

EXPECTED RETURN REQUIRED BY MARGINAL EQUITY INVESTORS

Cost of Equity

$$k_e = Rf + \beta \times \text{Equity Risk Premium}$$

Risk-free rate

Same currency and same real/nominal terms as cash flows.

Beta

Exposure to market risk, ideally estimated bottom-up from business mix and leverage.

Risk premium

Mature market premium plus country risk premium where relevant.

Risk rewarded in valuation is the risk that cannot be diversified away by the marginal investor.

HISTORICAL, IMPLIED, AND COUNTRY-RISK ADJUSTED

Equity Risk Premium

Approach	How it is estimated	Strength	Caution
Historical premium	Average past stock return over risk-free securities	Simple and familiar	Sensitive to period, T-bill vs T-bond, arithmetic vs geometric
Implied premium	Back-solve required return from current market level and expected cash flows	Forward-looking	Depends on market-level growth and payout assumptions
Country risk premium	Add default spread and/or volatility adjustment	Useful for emerging markets	Exposure depends on where the firm does business, not only where it is listed

Practical synthesis

Use implied mature-market premiums as the base; add country risk based on revenue exposure or operating exposure where relevant.

Beta: What Drives Market Risk?

Product / service cyclicality

Discretionary and cyclical businesses tend to have higher betas.

Operating leverage

High fixed costs amplify sensitivity of operating income to revenue shocks.

Financial leverage

More debt raises equity risk because interest is a fixed claim.

$$\beta^{L} = \beta^{U} \times [1 + (1 - t)\, D/E]$$

$$\beta^{Bu} = \Sigma \text{ segment weight} \times \beta^{U}(\text{segment})$$

Bottom-up beta is preferred when regression beta is noisy, outdated, or unavailable, especially for divisions and private firms.

Bottom-Up Beta: Teaching Recipe

VALUATION INSIGHTS FOR A BRIGHTER TOMORROW

1 **Define businesses**

Break the firm into operating segments.

2 **Find comparable firms**

Estimate unlevered beta by peer group.

3 **Weight by economics**

Use revenue, operating income, or value weights.

4 **Re-lever**

Apply the firm's target or current D/E ratio.

5 **Update over time**

Move beta toward stable business risk as the firm matures.

Why it is better

Lower standard error, reflects current business mix, reflects current/target leverage, works for private firms and divisions.

COUNTRY RISK IS ABOUT OPERATIONS, NOT HEADQUARTERS

Country Risk Exposure

$$E(\text{Return}) = R_f + \beta \times \text{Mature ERP} + \lambda \times \text{Country Risk Premium}$$

λ exposure factor

λ ≈ Domestic revenue share of firm / Domestic revenue share of average local firm

Approach	Assumption	Formula intuition
Equal exposure	All firms in country share country risk	Rf + CRP + β × ERP
Beta-scaled exposure	Country risk scales with market beta	Rf + β × (ERP + CRP)
Separate factor	Firms differ by revenue exposure	Rf + β × ERP + λ × CRP

Cost of Debt and WACC

Pre-tax k^d = Risk-free rate + Default spread

$$WACC = k_e \times E/(D+E) + k^d(1-t) \times D/(D+E)$$

Estimating k^d

Use traded bond yield, rating-based default spread, or synthetic rating from interest coverage.

Tax shield

Use marginal tax rate for forward-looking valuation; effective tax rates may transition toward marginal rates.

Market value weights

Weights should be based on market values, not book values. Use target weights when leverage changes.

Debt consistency The debt subtracted from firm value should be the same debt used to compute WACC.

PART 3

Cash Flows

From accounting numbers to valuation cash flows

DCF



Cash Flows

- In financial analysis, we are concerned about
  - **Cash flows to Equity**: These are the cash flows generated by the asset after all expenses and taxes, and also after payments due on the debt. Cash flows to equity, which are after cash flows to debt but prior to cash flows to equity
  - **Cash flow to Firm**: This cash flow is before debt payments but after operating expenses and taxes. This looks at not just the equity investor in the asset, but at the total cash flows generated by the asset for both the equity investor and the lender.
- These cash flow measures can be used to value assets, the firm's equity and the entire firm itself.

Measuring Cash Flows

To all claimholders: FCFF	To equity investors: FCFE
EBIT × (1 − tax rate)	Net income
− (Capital expenditures − Depreciation)	− (Capital expenditures − Depreciation)
− Change in non-cash working capital	− Change in non-cash working capital
= Free Cash Flow to Firm	− (Principal repaid − New debt issued)
	− Preferred dividends
	= Free Cash Flow to Equity

Interest tax shields are not added to FCFF; they are captured through after-tax cost of debt in WACC.

From Reported Earnings to Economic Earnings

Update

Use trailing 12 months and latest disclosures when annual reports are stale.

Normalize

Adjust cyclicality, restructuring, and abnormal earnings.

Cleanse

Remove financing items, non-recurring charges, and accounting distortions.

Capitalize

Treat operating leases and R&D as capital-like items when economically appropriate.

Base-year errors propagate into every forecast year and terminal value. Clean-up is not a cosmetic adjustment; it changes economics.

Free Cash Flow to Firm (FCFF)

$$FCFF = EBIT(1-t) - \text{Net CapEx} - \Delta \text{ Non-Cash WC}$$

Starts with operating income

FCFF is before debt payments. Do not subtract interest or principal.

Subtract reinvestment

Growth requires net capex and working capital investment.

Discount at WACC

FCFF belongs to both debt and equity claimholders.

From firm value to equity value

Operating asset value + cash/non-operating assets − debt − other claims = equity value.

CASH FLOW LEFT FOR EQUITY HOLDERS

Free Cash Flow to Equity (FCFE)

$$FCFE = \text{Net Income} - \text{Net CapEx} - \Delta \text{WC} - \text{Net Debt Repayment}$$

Stable leverage: $FCFE = NI - (1-\delta)(\text{Net CapEx} + \Delta \text{WC})$

Debt ratio δ

Represents the portion of reinvestment funded with debt.

Use when

Leverage is stable and the target is equity value, especially when dividends diverge from potential cash flow.

Caution More leverage can raise FCFE, but it also raises beta and cost of equity. It is not a free lunch.

Dividends vs FCFE

Use DDM when...	Use FCFE model when...
Dividends and buybacks are close to FCFE over time	Dividends are much lower or much higher than FCFE
FCFE is hard to estimate, e.g., banks and financial services	Dividends are unavailable, e.g., private firms or IPOs
Management payout policy reflects true capacity	Management smooths dividends or hoards cash
Regulated and mature equity business	Non-financial firms with observable reinvestment needs

Rule of thumb from Damodaran: use FCFE if dividends are below 80% of FCFE or above 110% of FCFE over a meaningful period.

Reinvestment: Net CapEx and Working Capital

Net capital expenditures

Net CapEx = Capital expenditures − Depreciation. High-growth firms generally need higher net capex.

Non-cash working capital

Use non-cash current assets minus non-debt current liabilities. Increases reduce cash flow.

Forecasting method

Forecast working capital as a percent of revenues, not as volatile annual changes.

Reinvestment = Net CapEx + Δ Non-Cash WC

Sales / Capital Ratio = Δ Revenue / Reinvestment

A terminal value with positive perpetual growth and zero reinvestment is usually internally inconsistent.

Accounting Adjustments That Matter

Issue	Valuation treatment	Why it matters
Operating leases	Convert to debt; adjust operating income	A financing commitment hidden as operating expense
R&D expenses	Capitalize and amortize as research asset	Creates future benefits; important for tech/pharma
One-time charges	Remove if genuinely non-recurring; normalize if recurring pattern	Avoid building temporary noise into perpetuity
Tax rates / NOLs	Use marginal tax rate eventually; apply NOLs in early years	Taxes affect FCFF, WACC, and terminal value

The goal is not to “fix accounting” universally; it is to restate accounting into economic cash-flow logic for valuation.

PART 4

Growth & Reinvestment

Where growth comes from and when it creates value

DCF



DO NOT EXTRAPOLATE BLINDLY

Three Ways to Estimate Growth

Historical growth

Useful starting point, but sensitive to period, averaging method, size, cyclicality, and negative base earnings.

Analyst forecasts

Useful information, especially near term, but forecasts can be correlated and biased by access to management.

Fundamental growth

Best anchor: growth comes from reinvestment and returns on that reinvestment.

$$\text{Growth} \approx \text{Reinvestment Rate} \times \text{Return on Capital}$$

Growth Patterns in DCF Valuation

How growth is modeled and what determines the length and intensity of high growth

1 Three Common Growth Pattern Assumptions

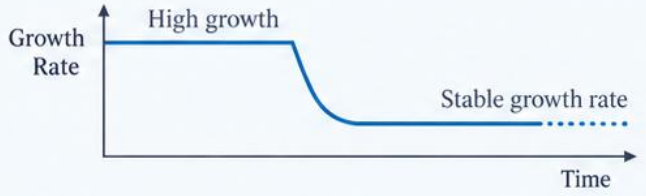
Stable Growth (1-stage)

No high-growth phase; the firm is already in stable growth.



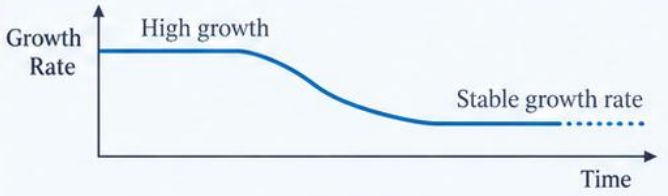
High Growth then Stable Growth (2-stage)

A period of high growth followed by a drop to a stable long-run growth rate.



High Growth with Gradual Transition (3-stage / n-stage)

A period of high growth followed by a gradual decline toward stable growth; margins and growth can vary year by year.



2 Determinants of Growth Patterns

Size of the Firm

As firms become larger, it becomes harder to sustain very high growth rates.

Current Growth Rate

Higher current growth often signals higher future growth and a longer high-growth period, though past growth alone is not fully reliable.

Barriers to Entry & Differential Advantages

Sustained high growth depends on strong competitive advantages and barriers to entry; the stronger and longer-lasting they are, the longer high growth can persist.

Teaching Insight | In DCF, the analyst must choose both the pattern of growth and the economic reasons why that pattern can be sustained.

Fundamental Growth: The Engine

Investment in new projects

How much of current earnings/cash flow is reinvested?

→

Return on investment

How much operating income or equity income is generated by reinvestment?

→

Earnings growth

Sustainable growth emerges from the product of reinvestment and return.

$$g^{EPs} = \text{Retention Ratio} \times ROE$$

$$g^{EBIT} = \text{Reinvestment Rate} \times ROC$$

Equity Growth Formulas

Situation	Formula	Interpretation
Stable ROE	$gEPS = b \times ROE$	Retention ratio drives reinvestment; ROE drives quality
Changing ROE	$gEPS = b \times ROE_{t+1} + (ROE_{t+1} - ROE_t)/ROE_t$	Growth includes improvement in existing assets
Net income / FCFE model	Equity reinvestment rate × ROE	Use real reinvestment instead of retained earnings
Long-term constraint	$g \leq ROE$	No firm can grow earnings per share above ROE forever

Caution Growth from improving ROE is transitional. In the long run, competition limits excess returns.

FOR FCFF MODELS

Operating Income Growth Formulas

Reinvestment Rate = (Net CapEx + Δ WC) / EBIT(1−t)

ROC = EBIT(1−t) / Book Capital

Situation	Growth formula	Teaching point
Stable ROC	gEBIT = Reinvestment Rate × ROC	Quality of investment reduces required reinvestment for a given growth
Changing ROC	$g = ROC_{t+1} \times \text{Reinvestment} + (ROC_{t+1} - ROC_t)/ROC_t$	Improving existing assets adds temporary growth
Multiple-year ROC change	Spread improvement component across years	Avoid one-year spikes that distort value

Negative Earnings and Changing Margins

Revenue growth path

Start with current revenue; fade growth as scale increases.

→

Target margins

Move operating margin toward a sustainable industry or business-model benchmark.

→

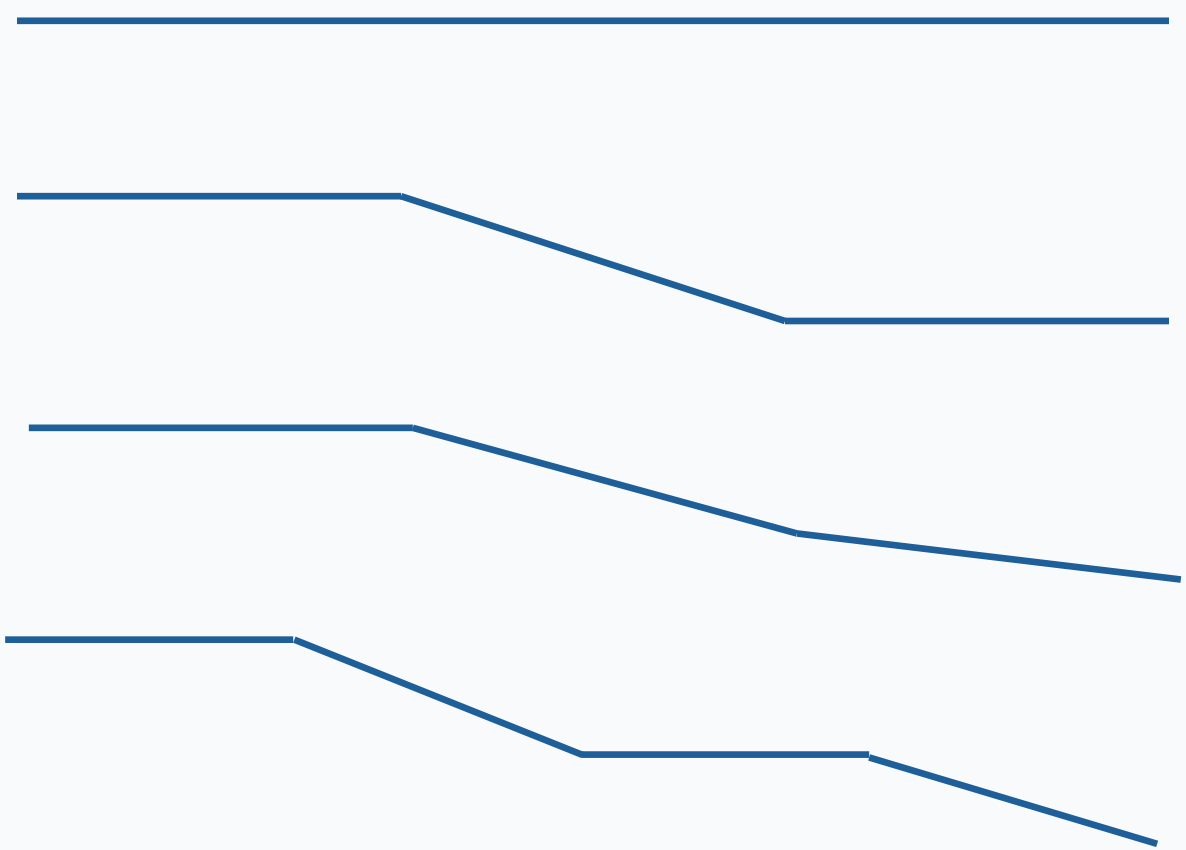
Reinvestment needs

Use sales-to-capital ratio to estimate reinvestment needed for revenue growth.

$$FCFF = Revenue \times Operating\ Margin \times (1-t) - Reinvestment$$

This is the natural structure for platform, e-commerce, AI, and early-stage digital firms.

Growth Patterns



Stable

Already mature; no high-growth phase.

Two-stage

High growth drops to stable growth.

Three-stage

High growth fades through transition.

n-stage

Every year can differ: margins, risk, growth.

Determinants: firm size, current growth, barriers to entry, competitive advantage, and how long excess returns can last.

TERMINAL ASSUMPTIONS SHOULD CONVERGE

Stable Growth Must Look Stable

Dimension	High-growth firm	Stable-growth firm
Risk	High beta / high uncertainty	Average or mature-business risk
Reinvestment	High net capex and working capital needs	Lower and sustainable reinvestment
Returns	High ROC / ROE if competitive advantage exists	ROC closer to WACC; ROE closer to k_e
Leverage	Often low or changing	Closer to industry or target leverage
Payout	Low dividends / reinvestment priority	Higher payout; reinvestment tied to g/return

Stable growth reinvestment logic Reinvestment Rate = Stable Growth / Stable ROC. Payout Ratio = 1 − Stable Growth / Stable ROE.

PART 5

Terminal Value

DCF

Getting closure without abusing perpetuity



PUBLIC FIRMS MAY HAVE VERY LONG LIVES

Why Terminal Value Exists

Value = PV(explicit forecast cash flows) + PV(terminal value)

The problem

We cannot forecast annual cash flows forever with meaningful detail.

The solution

Forecast a growth period explicitly, then value all cash flows beyond the horizon as terminal value.

The danger

Terminal value often dominates DCF, so small mistakes in stable assumptions have large valuation effects.

Terminal value is not a plug. It is the value of a stable company after transition.

CHOOSE CLOSURE CAREFULLY

Ways to Estimate Terminal Value

Method	Best use	Limitation
Liquidation value	Assets are separable and marketable	Not appropriate for most going concerns
Exit multiple	Quick market-based triangulation	Turns the DCF into relative valuation
Stable growth model	Going-concern valuation with mature steady state	Requires judgment on stable g, risk, reinvestment, and excess returns

$$\text{Terminal Value}_n = CF_{n+1} / (r - g)$$

POWERFUL BUT FRAGILE

Stable Growth Formula

$$TV_n = FCFF_{n+1} / (WACC - g) \quad \text{or} \quad TV_n = FCFE_{n+1} / (k_e - g)$$

Constraint 1

Stable growth cannot exceed long-term growth of the economy in the valuation currency.

Constraint 2

Stable reinvestment must support stable growth: Reinvestment = g / ROC.

Constraint 3

Stable risk, leverage, and margins must look like a mature company, not a high-growth startup.

Most common TV error

Assuming high perpetual growth, low reinvestment, and high excess returns simultaneously.

BEFORE TRUSTING THE VALUATION

Terminal Value Sanity Checks

Check	Question to ask
Growth ceiling	Is stable g below economy growth in the currency used?
Reinvestment consistency	Does reinvestment rate equal g/ROC or g/ROE?
Excess returns	Is ROC > WACC defensible forever, or should competition erode it?
Capital intensity	Can sales grow with the assumed sales-to-capital ratio?
Share of value	Is too much value coming from terminal value? Why?
Sensitivity	How much does value change if WACC or g moves by 50–100 bps?

DCF SHOULD PRODUCE A VALUE RANGE

Sensitivity Is Not Optional

Scenario	Stable g	WACC / k_e	ROC / ROE	Implication
Bear	Lower	Higher	Lower	Lower terminal value and lower explicit cash flows
Base	Defensible	Consistent	Industry-normal	Reference case
Bull	Higher but capped	Lower	Higher but defendable	Value upside must have an economic narrative

Teaching frame

Do not ask whether a DCF is “right”. Ask which assumptions the market price is implying and whether those assumptions are plausible.

PART 6

Model Selection & Application

DCF

Choosing the right DCF for the asset



DCF Model Selection Guide

How to choose the right cash flow, discount rate, and growth pattern

1 Which cash flow should I discount?

Rule: match the cash flow to the object being valued.

Use Equity Valuation

- Firm has stable leverage
- You are valuing equity / stock

Discount: Cash flows to equity (FCFE or dividends)

Use Firm Valuation

- Leverage is too high or too low and expected to change over time
- Leverage information is incomplete
- You are more interested in valuing the firm than only the equity

Discount: Cash flows to firm (FCFF)

2 If using cash flows to equity: Dividends or FCFE?

Use Dividend Discount Model

- Dividends and buybacks are close to FCFE over time
- FCFE is difficult to estimate (e.g., banks and financial service firms)

Dividends (buybacks)
Cash returned to shareholders

VS.

FCFE
Cash available to equity holders after reinvestment

Use FCFE Model

- Dividends are materially different from FCFE
- Practical rule of thumb: if dividends are below 80% of FCFE or above 110% of FCFE over 5 years, prefer FCFE
- Use when dividends are unavailable (e.g., private companies, IPOs)

3 What discount rate should I use?

Cash flow ↔ discount rate match

Dividends / FCFE	→ Cost of Equity
FCFF	→ Cost of Capital (WACC)

Currency match

Estimate the risk-free rate in the same currency as the cash flows.

Real vs nominal

Real cash flows	→ Real discount rate
Nominal cash flows	→ Nominal discount rate

- **Low inflation** (< 10%): usually stay with nominal cash flows (since taxes are based on nominal income)
- High inflation (> 10%): real cash flow valuation may be more appropriate

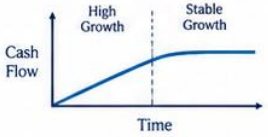
4 Which growth pattern should I use?

Stable Growth Model



- Large, mature firm
- Growth close to or below the economy's growth rate
- Regulation or business characteristics constrain extraordinary growth
- Risk and reinvestment are already near stable levels

2-Stage Growth Model



- Large firm growing at a moderate rate
- One main product or competitive advantage with a finite life
- High growth for a period, then shift to stable growth

3-Stage / n-Stage Model



- Small or young firm with very high growth
- Strong barriers to entry or unusual characteristics
- Growth fades gradually rather than all at once

Teaching Insight | In DCF, good valuation comes from matching the cash flow definition, discount rate, and growth structure to the economics of the business.

Which Cash Flow Should I Discount?

Use equity valuation when...	Use firm valuation when...
Leverage is stable or expected to stay stable	Leverage is too high, too low, or expected to change
You are valuing common equity directly	You are valuing the entire business or enterprise value
Debt payments and new debt issues are estimable	Debt payments/issues are volatile or partial leverage data is available
FCFE or dividends are reliable	FCFF is more transparent than FCFE

Core decision The chosen cash flow determines the required discount rate and the output value.

Which Growth Pattern Should I Use?

Company profile	Suggested model	Rationale
Large, mature, regulated, or economy-like growth	Stable growth	Already resembles a steady-state firm
Large firm with moderate growth or finite advantage	Two-stage	High growth for a period, then stable
Small firm with high growth or strong barriers to entry	Three-stage	High growth fades gradually
Digital firm, negative earnings, changing margins	n-stage FCFF	Revenue, margins, risk, and reinvestment vary year by year

The Building Blocks of Valuation

Choose cash flow

Dividends
FCFE
FCFF

→

Choose discount rate

Cost of equity
WACC
Real/nominal/currency matched

→

Choose growth pattern

Stable
Two-stage
Three-stage
n-stage

Intrinsic Value = Present Value of correctly matched cash flows

Loose Ends After Operating Valuation

Item	Treatment
Cash and marketable securities	Keep out of operating cash flows; add back after valuing operating assets
Holdings in other firms	Value separately and add the proportion owned; avoid double counting
Debt and non-equity claims	Subtract claims consistent with WACC calculation
Convertible debt / warrants / options	Separate debt and equity-option components; subtract option value before per-share value
Number of shares	Divide remaining common equity value by shares outstanding

Value per share = (Equity value − Value of non-stock equity claims) / Shares outstanding

Special Situation: Distress

Distress-adjusted equity value = DCF equity value × P(survival) + Distress sale equity value × P(distress)

When needed

The firm has a meaningful probability of failing before reaching stable growth.

Estimate probability

Use bond ratings, market debt prices, or statistical distress models.

Estimate distress value

Usually based on a discount to book capital or asset sale values.

A normal DCF may overstate value if survival is implicitly assumed but not credible.

DCF for Digital and AI Businesses

Challenge	Modeling response
Negative earnings	Forecast revenues and margins explicitly; do not compute meaningless earnings growth
Intangible reinvestment	Consider capitalization logic for R&D, platform development, data, and customer acquisition
Changing risk profile	Allow beta, cost of debt, and leverage to migrate as business matures
Network effects and optionality	DCF base case plus scenarios; real-option thinking for expansion opportunities
High terminal value dependence	Use strict stable-growth and reinvestment discipline

Key teaching point

For digital firms, the DCF is not less relevant; it simply forces explicit stories about scale, margins, reinvestment, and risk.

Mini Case Pattern: Amazon-Style n-Stage DCF

Current state

Low or negative margin
High revenue growth
High uncertainty

→

Transition

Growth fades
Margins improve
Reinvestment remains high

→

Stable state

Mature beta
Stable debt ratio
Reinvestment = g/ROC

Value = PV(negative early FCFF + improving mid-stage FCFF + disciplined terminal value)

Common DCF Mistakes

Mistake	Why it is dangerous
Mismatching cash flows and rates	Produces biased values even when arithmetic is correct
Using book weights in WACC	Financing costs are priced by markets, not accounting book values
Perpetual growth above economy growth	Implies the firm eventually dominates the economy
Positive perpetual growth with zero reinvestment	Growth requires capital unless returns are infinite
Ignoring options, leases, cash, or cross-holdings	Creates double counting or missing claims
Using point estimates only	Hides uncertainty and narrative dependence

Classroom Exercise

Task

Choose a listed digital business or platform company and build a one-page DCF narrative before opening Excel.

Question	Output
What cash flow will you discount?	FCFE, FCFF, or dividends; justify claimholder perspective
What are the key forecast drivers?	Revenue growth, margin path, reinvestment, working capital
What risk profile applies?	Beta, country exposure, cost of debt, target leverage
When does the firm become stable?	Growth fade, stable margins, stable risk, stable reinvestment
What market assumptions are implied?	Reverse-engineer price-implied growth or margin

Ten Takeaways

#	Takeaway
1	DCF values expected future cash flows, not accounting earnings.
2	The discount rate prices risk and must match the cash flow claimholder.
3	Equity valuation and firm valuation can converge when done consistently.
4	Growth is valuable only when returns exceed the cost of capital.
5	Reinvestment is the cost of growth.
6	Terminal value must reflect a mature firm, not a fantasy firm.
7	Digital firms require explicit revenue, margin, and reinvestment paths.
8	Loose ends can materially affect per-share value.
9	Sensitivity and reverse valuation are part of the valuation, not appendices.
10	The best valuation is transparent about uncertainty.

References

Aswath Damodaran, Discounted Cash Flow Valuation teaching material.
Aswath Damodaran, Investment Valuation: Tools and Techniques for Determining the Value of Any Asset, 3rd ed.
Damodaran Online: datasets, spreadsheets, and lecture notes for valuation.
Recommended practice: use this deck with a spreadsheet exercise and a reverse-DCF discussion.

Thank You · Questions & Discussion

[Oktofa Yudha Sudrajad] · Digital Business Valuation · SBM ITB

REFERENCES

Damodaran, A. (2012). Investment Valuation: Tools and Techniques for Determining the Value of Any Asset, 3rd Edition (University Edition). Hoboken, NJ: John Wiley & Sons. Chapters 2-9

Companion datasets and valuation spreadsheets: damodaran.com

Thank You

Questions & Discussion

